MINIFLY RESEARCH / DESIGN PROPOSAL / 01

Does byte pretraining transfer?

Run two linked tests with the same V9.2 checkpoints. First, measure whether more unlabelled prose lowers frozen prediction loss on unseen books. Then measure whether those checkpoints improve QA when every arm receives the same amount of teaching. Keep the two decisions separate.

Reference and scope

V9.2 uses fixed hashed suffix features, a 256-byte output distribution and local next-byte updates. Its mutable predictor arrays occupy about 2.75 MiB. The earlier matched comparison on one Alice suffix favored V9.2 over Round 8, 2.757 versus 2.857 bits per byte. That comparison supports choosing a reference; it does not measure a dose curve or cross-book transfer.

Phase A / pretraining arms

0 MiB. Save the initialized predictor. **1 MiB and 10 MiB.** Start from that same initialization and save checkpoints after exactly 1 and 10 MiB of one causal pass. The 1 MiB stream is a prefix of the 10 MiB stream. **10 MiB shuffled.** Apply a seeded permutation within each consumed book segment, preserving segment lengths, byte counts, boundary resets and update count. This tests whether prose order contributes beyond exposure to byte frequencies.

Use three fixed training book orders, with matched intact and shuffled arms. The 0 MiB model is identical across those orders. Keep the hash implementation, model dimensions, update rates and numeric precision fixed. One MiB means 1,048,576 raw bytes after normalization. Never carry input history across books or into downstream QA. Preserve weights and bias only when transferring the predictor.

What qualifies as success

A positive Phase A result supports more data for this predictor. A positive Phase B result supports transfer through the current QA fallback. Failure to improve QA with a better predictor localizes the limit to the tested interface or the availability of useful fallback opportunities. A larger store, a shared representation, QA2 and structural-map pretraining are outside this first experiment.

Prose protocol and decisions

Freeze corpus membership before scoring

Assemble public-domain prose with recorded source, author, edition and SHA-256. Strip publisher headers and footers, normalize line endings, preserve UTF-8, and deduplicate across splits. Assign by author and work, including duplicate editions, before selecting any bytes. The training pool must support each 10 MiB stream without repeating a byte segment. Reserve development authors for implementation checks and at least eight distinct evaluation authors with one or more untouched books each.

Publish a manifest with raw and normalized hashes, split membership, byte spans, order seeds and boundary rules. Check cross-split overlap using exact and near-duplicate text checks. Keep evaluation authors closed until the code, manifests and thresholds are frozen. The proposed sizes and thresholds are development design choices.

Measure frozen next-byte prediction

At each byte, score the model distribution using only earlier bytes in that book. Supply the observed byte as history after scoring. Disable all parameter writes and verify weight and bias hashes before and after evaluation. Reset history at book boundaries. Report total negative log probability divided by evaluated raw bytes in bits per byte, and report each book as well as author-macro and byte-weighted means.

Include the initialized model, an empirical byte-frequency predictor and a train-only Markov-3 reference fitted to the same 10 MiB intact data. Report intact versus shuffled and 10 versus 1 MiB. Average the three training orders within each evaluation author for paired intervals; do not count orders as new authors.

Proposed Phase A screen

At 10 MiB, require at least 0.05 bits per byte improvement over 1 MiB and at least 0.05 over the shuffled 10 MiB control, on the author-macro score, with paired 95% intervals excluding zero. Report the margin to Markov-3 independently. If 1 MiB helps and 10 MiB adds little, the dose curve has flattened over this range. If the gain is confined to one author, inspect overlap and genre mix before increasing the dose.

QA transfer through the predictor fallback

Use the same QA interface in every arm

V205 QA1 uses the Round 8 predictor as its fallback. Substitute V9.2 in all transfer arms, including the zero-pretraining arm. Leave the V205 factual address, full-strength write policy, learned gate, strict frozen question-frequency weighting and answer-step memory unchanged. The original V205 with Round 8 is an engineering reference for the adapter. Do not interpret a comparison between architectures as a pretraining effect.

Pretrained state enters QA only as the V9.2 weight and bias checkpoint. Clear history. Initialize all specialist state identically, then give all arms nested teaching sets of 50, 200 and 800 complete questions. Use identical stories, order, cues and observed feedback within each paired world. In the main test, continue predictor learning during QA teaching with identical rules. At the 200-question budget add a frozen-predictor contrast for 0 and 10 MiB to distinguish carried skill from adaptation during teaching.

Fresh test with room to improve

Seal 12 paired story batches with separate original and unseen statement renderings. Include unseen names and changed location prefixes. Keep questions and answer targets paired across renderings. Use a development-only check to ensure that the proposed 200-question primary condition has measurable fallback use and room for improvement. If it has neither, report the interface as uninformative for transfer before opening the sealed set.

At evaluation freeze predictor, gate, answer memory and question-frequency statistics. Let episodic memory write permitted story statements. Provide the fixed query cue, score the model-generated answer and its own newline, and feed no gold answer bytes after scoring. The primary endpoint is exact answer plus newline on unseen renderings at 200 questions.

Attribute any gains

Log the source of each emitted byte and score first-byte fallback, continuation fallback, specialist-only paths, newline completion and gate hit and false-trigger rates. Use an evaluator-only teacher-forced diagnostic with common prefixes to compare fallback heads directly. Verify specialist state digests are identical across pretraining arms at matched points. Generated answers stay outside the learning stream. Gate improvements require a distinct coupling experiment.

Promotion rules and execution

Proposed Phase B screen

Require at least +5 percentage points over the 0 MiB V9.2 arm on the primary endpoint, with a batch-paired 95% interval excluding zero and no more than a 3-point mean drop on original wording. Compare against shuffled pretraining and report the exact teaching doses at which each arm reaches the same accuracy. If 10 MiB reaches a fixed accuracy with fewer labels, quantify the savings using only measured budgets.

Use story batches as the paired units, averaging shared predictor orders within each batch. Report each rendering, the full teaching curve and actual fallback denominators. Report Phase A and Phase B separately even if only one passes. A prose gain alone justifies a larger prose study if the curve is still improving. A QA gain supports scaling this transfer interface. If fallback usage is sparse or prose improves without QA benefit, inspect the byte-level routing before changing the model.

Engineering qualification before final runs

Cycle 1: match the V9.2 numerical reference, verify predict-before-update, corpus disjointness, exact byte budgets, history reset and frozen evaluation. Cycle 2: run all arms on development text and complete QA lifetimes. Check equal teaching streams, no gold answer ingress at evaluation, identical specialist state and output attribution. Cycle 3: restore a persisted predictor checkpoint in a fresh process and compare weights, stream position, loss and generated QA bytes with an uninterrupted run. Verify corruption rejection and deterministic execution.

Freeze source hashes, corpus and story manifests, orders, comparator definitions, thresholds and analysis. Record wall and CPU time, peak RSS, checkpoint bytes and throughput at each dose. Use streaming updates and bounded caches. The 10 MiB cap is a first development ceiling; inspect measured cost and both dose curves before authorizing a larger run.

Source records

V9.2 predictor design and V201 byte runtime. Matched prose comparison from 6a96c087. V205 QA1 learner and results from 0b84f086. This document proposes new work. No training or evaluation has been launched.